Exercise 5.24 Part 2

2. $\set{12n + 9m : n, m \in \mathbb{Z}} = \set{3k : k \in \mathbb{Z}}$.

Proof.

Denote $A = \set{12n + 9m : n, m \in \mathbb{Z}}$ and $B = \set{3k : k \in \mathbb{Z}}$.

($\subseteq$) By Fact **2.3**, we know for all integers x, y, then $4x + 3y$ is an integer. Now B has a subset $\set{3(4x + 3y) : k = 4x + 3y}$. Notice that such a subset is the same set as A. Thus, $A \subseteq B$.

($\supseteq$) We know that -1 times an integer is also an integer. Now A has a subset $\set{12n -9n : m = -n}$. Notice that such a subset is the same set as B. Thus, $B \subseteq A$.

Now $A \subseteq B$ and $B \subseteq A$. Therefore, by Definition **5.19**, then $\set{12n + 9m : n, m \in \mathbb{Z}} = \set{3k : k \in \mathbb{Z}}$.

■

Exercise 5.24 Part 3

3. $\set{5j + 1 : j \in \mathbb{Z}} = \set{5j + 6 : j \in \mathbb{Z}}$.

Denote $A = \set{5j + 1 : j \in \mathbb{Z}}$ and $B = \set{5j + 6 : j \in \mathbb{Z}}$.

($\supseteq$) By Fact **2.3**, we know for all integers x, then $x + 1$ is an integer. Now A has a subset $\set{5(x + 1) + 1 : j = x + 1}$. Notice that such a subset is the same as B. Thus, $B \subseteq A$.

($\subseteq$) We know for all integers y, then $y - 1$ is an integer. Now B has a subset $\set{5(y - 1) + 6 : j = y - 1}$. Notice that such a subset is the same as A. Thus, $A \subseteq B$.

Now $A \subseteq B$ and $B \subseteq A$. Therefore, by Definition **5.19**, then $\set{5j + 1 : j \in \mathbb{Z}} = \set{5j + 6 : j \in \mathbb{Z}}$.

■